

ω Cen: the mass-modelling procedure

oCen_dm project notes

2026 September 21

Abstract

The companion note describes how the kinematic data were prepared. This one describes what is done with them: the mass model, the dynamical solver, the likelihood, the priors, and the inference. The question is whether ω Cen requires mass beyond its stars, remnants and any central black hole, and it is posed as a comparison of Bayesian evidence between two model families that differ by one component. Everything here is deliberately low-dimensional: the data are 89 binned points over roughly 0.1–55 pc. Three isotropic fits without instrument scales are complete. They have no persuasive halo preference and all leave structured residuals, so the posteriors are conditional baselines. This note describes the implemented machinery and distinguishes it from the proposed anisotropy extension.

1 The question, and how it is posed

Two families are fitted to the same observations and compared by evidence.

K1, no dark matter stars, a compact remnant population, and a central point mass.

K2, dark matter the same three components plus a truncated generalized NFW halo.

K1 is the zero-halo limit of the K2 equations. Zero is outside the adopted log-uniform K2 mass prior, so it is not a sampled nested point. The Bayes factor $\ln Z_{K2} - \ln Z_{K1}$ compares the separately specified families and their priors. Two K2 variants are run separately, a cored halo ($\gamma = 0$) and a cuspy one ($\gamma = 1$), rather than fitting γ : making the inner slope free gives the halo shape freedom the data do not support, and the two limiting cases bracket it.

2 The mass model

The potential is spherical and the components add:

$$M(< r) = M_{\star}(< r) + M_{\text{rem}}(< r) + M_{\bullet} + M_{\text{DM}}(< r). \quad (1)$$

Stars. A multi-Gaussian expansion, deprojected analytically under spherical symmetry, scaled by a single free total mass $M_{\star}$. Using an MGE rather than a Plummer sphere matters because the tracer density appears in the Jeans equation directly, so its shape is not a free choice.

The default profile is not the surface brightness alone. It is a *composite*: HST star counts inside 25'' spliced onto the Trager surface-brightness profile outside, because the light profile is unreliable in the crowded core where a handful of bright giants dominate it. The Trager-only alternative is selectable. The same shape supplies both the tracer density $\nu(r)$ and the stellar mass density, which assumes a radius-independent mass-to-light ratio; mass segregation makes that an approximation.

Remnants. A Plummer sphere with free mass M_{rem} and scale a_{rem} . It is a separate component rather than an inflation of the stellar mass-to-light ratio so that its mass appears under its own name in every posterior and can be confronted with multi-mass predictions later. The parametrisation is phenomenological: the point is to let a centrally concentrated dark component of *known* astrophysical origin compete with the halo, so that a halo detection cannot be manufactured by forbidding one.

Central point mass. $M_{\bullet}$, an intermediate-mass black hole or any unresolved central mass.

Dark halo (K2 only). A truncated generalized NFW profile,

$$\rho(r) = \rho_s \left(\frac{r}{r_s} \right)^{-\gamma} \left(1 + \frac{r}{r_s} \right)^{\gamma-3} T(r), \quad T(r) = \left[1 + \left(\frac{r}{r_t} \right)^2 \right]^{-2}. \quad (2)$$

The taper falls as r^{-4} outside r_t , giving an outer density proportional to r^{-7} and finite total mass. The fitted family fixes $r_t = 1000$ pc, outside the measured radii. This is a phenomenological taper, not a tidal boundary derived from the cluster’s orbit; it should not be confused with the much smaller radii in the survival experiments.

The halo is parametrised by $M_{\text{DM}}(< 100 \text{ pc})$ and r_s rather than by ρ_s and r_s . The normalisation is solved from the enclosed mass at 100 pc. Both this mass and r_s have log-uniform priors. The 100-pc mass remains an extrapolation beyond the outermost measured radius; the scale radius is poorly constrained in the baseline runs.

3 Anisotropy

The implemented choices are isotropic, constant β , or the monotonic form

$$\beta(r) = \beta_0 + (\beta_{\infty} - \beta_0) \frac{r^2}{r^2 + r_{\beta}^2}. \quad (3)$$

The last has three parameters and a closed-form Jeans integrating factor. It is the logistic form passed to JamPy with $\alpha = 2$, although the two engines still use different numerical approximations. It can cross zero once; it cannot have an intermediate maximum. A turnover extension is motivated by the residuals and remains unimplemented.

For a cored tracer in a central point-mass potential, the necessary condition from [An & Evans \(2006\)](#) is $\beta(0) \leq -1/2$. The varying family imposes $\beta_0 \in [-1, -1/2]$. This is necessary, not sufficient, for a non-negative distribution function. The condition also applies to constant anisotropy: the isotropic baseline and constant- β prior $[-1, 0.5]$ deliberately allow moment models that violate it. Their purpose is diagnostic, and their central-mass estimates must be interpreted accordingly.

4 The dynamical solver

Given $\nu(r)$ from the tracer MGE, $M(< r)$ from the mass model and $\beta(r)$, the spherical Jeans equation

$$\frac{d(\nu\sigma_r^2)}{dr} + \frac{2\beta\nu\sigma_r^2}{r} = -\frac{\nu GM(< r)}{r^2} \quad (4)$$

integrates with the factor $g(r) = \exp(2 \int \beta/t dt)$ to

$$\nu\sigma_r^2(r) = \frac{1}{g(r)} \int_r^{\infty} g(s) \nu(s) \frac{GM(< s)}{s^2} ds, \quad (5)$$

and the projected second moments follow (Binney & Mamon, 1982):

$$\Sigma\sigma_{\text{los}}^2(R) = 2 \int_R^\infty \left[1 - \beta \frac{R^2}{r^2}\right] \nu\sigma_r^2 \frac{r \, dr}{\sqrt{r^2 - R^2}}, \quad (6)$$

$$\Sigma\sigma_{\text{pmR}}^2(R) = 2 \int_R^\infty \left[1 - \beta + \beta \frac{R^2}{r^2}\right] \nu\sigma_r^2 \frac{r \, dr}{\sqrt{r^2 - R^2}}, \quad (7)$$

$$\Sigma\sigma_{\text{pmT}}^2(R) = 2 \int_R^\infty [1 - \beta] \nu\sigma_r^2 \frac{r \, dr}{\sqrt{r^2 - R^2}}. \quad (8)$$

All three projections and $\Sigma(R)$ share their quadrature nodes, so one model evaluation delivers the line-of-sight, radial and tangential predictions together.

Numerics. The substitution $r = R \cosh u$ removes the integrable singularity at $r = R$ exactly, since $r \, dr / \sqrt{r^2 - R^2} = r \, du$, leaving smooth integrals that fixed Gauss–Legendre nodes handle to high accuracy. $\nu\sigma_r^2$ is tabulated once per model on a 600-point logarithmic grid and splined in the log. A full set of projected profiles costs a few milliseconds. The completed rung-0 runs required 3.2–14.3 million likelihood calls and 4.0–20.6 hours; sampling efficiency is a practical limitation.

One defect worth recording: the outer integral was originally evaluated as cumulative[−1] – cumulative, which loses all significance once the remaining integral is small, and σ_r collapsed to zero around three MGE scale lengths. It is now accumulated in reverse. The runs made before that fix are quarantined.

5 The likelihood

Each dataset is a set of binned dispersions with asymmetric errors. The log-likelihood is a split normal,

$$\ln \mathcal{L} = \sum_i -\frac{1}{2} \left(\frac{m_i - d_i}{\varepsilon_i} \right)^2 + \ln \frac{\sqrt{2/\pi}}{\varepsilon_i^- + \varepsilon_i^+}, \quad \varepsilon_i = \begin{cases} \varepsilon_i^+ & m_i > d_i \\ \varepsilon_i^- & \text{else,} \end{cases} \quad (9)$$

summed over datasets, which are treated as independent. The normalisation is *shared* between the two sides, which is what makes the density continuous at $m_i = d_i$ and integrate to one; giving each side its own $-\ln \varepsilon$, as this did until 2026-09-20, produced a jump of $\varepsilon^+/\varepsilon^-$ across every datum.

Three things happen between the model and the datum.

Radial averaging. For the datasets we measured ourselves — the four HST and *Gaia* profiles — the model is averaged over the radii of the stars actually measured in each bin, eight equal-count quantiles stored with the data, rather than over a complete annulus with $\Sigma(R)R$ weighting. Where the selection’s coverage changes inside a bin the two differ: HST’s 300–340'' sample has a star-weighted mean radius of 313.2'' against the annulus midpoint of 320'', worth up to 1.17σ . The 29 **MUSE** bins are published, carry no such information, and are still averaged over complete annuli.

Streaming. The model predicts the full second moment; the data are dispersions about a rotating mean. Each dataset that needs it carries $\langle \bar{v}^2 \rangle$, and the comparison is with $\sqrt{\sigma_{\text{model}}^2 - \langle \bar{v}^2 \rangle}$. The uncertainty on that term is propagated into the datum’s error for the HST and *Gaia* profiles. **It is not for MUSE**, whose rotation and dispersion were fitted jointly by the survey and would need their covariance rather than an independent error: at the innermost bin the first-order rotation contribution is about 1.22 km s^{-1} against quoted errors of $1.90/2.52 \text{ km s}^{-1}$, so this is not negligible and is an outstanding item.

Instrument scales. Optional multiplicative nuisances s_{inst} on the model prediction, one per instrument, with uniform priors. They absorb a constant calibration offset. They are also strongly degenerate with the halo mass, which is why the adopted family comparisons use **-no-scales**. A separate scale-freeing diagnostic may be run after the sequence, but its evidence is not compared. The bare CLI still creates the three legacy scale parameters; the explicit flag is essential.

6 Parameters and priors

Parameter	Prior	Unit	Meaning
$M_{\star}$	log-uniform $[10^6, 10^7]$	$M_{\odot}$	stellar mass
M_{rem}	log-uniform $[10^4, 3 \times 10^6]$	$M_{\odot}$	remnant mass
a_{rem}	log-uniform $[0.3, 20]$	pc	remnant scale radius
$M_{\bullet}$	log-uniform $[10^2, 3 \times 10^5]$	$M_{\odot}$	central point mass
β_0	uniform $[-1, -1/2]$	—	central anisotropy (An & Evans)
β_{∞}	uniform $[-1, 1]$	—	outer anisotropy
r_{β}	log-uniform $[0.5, 100]$	pc	anisotropy scale
$M_{\text{DM}}(< 100 \text{ pc})$	log-uniform $[10^3, 3 \times 10^7]$	$M_{\odot}$	K2 only
r_s	log-uniform $[3, 1000]$	pc	K2 only
s_{inst}	uniform, $\sim [0.7, 1.3]$	—	optional, one per instrument
D	normal (5.43, 0.05)	kpc	free in the adopted runs

Table 1: Available parameters. Rung 0 fixes anisotropy and removes instrument scales, leaving 5/7 parameters for K1/K2, including distance. Constant anisotropy gives 6/8; the implemented varying form gives 8/10. MUSE’s diagnostic scale prior is $[0.85, 1.15]$; the Gaia priors are $[0.7, 1.3]$.

7 Inference

Nested sampling with **ultranest**: 400 live points, $d \ln Z < 0.5$, MLFriends by default with a slice stepper available for the higher-dimensional variants. Posterior mass profiles and residuals are the main scientific products; evidence compares the stated families after their adequacy and input comparability are checked.

Each run writes an equally weighted posterior, a summary with evidence and per-dataset χ^2 at the best sampled point, a subset of radial mass profiles, and **run.yaml**. The quoted evidence errors are sampler estimates; they do not include uncertainty from model inadequacy or missing data covariance.

New runs save the actual likelihood arrays and resolved model before sampling: MGE coefficients, priors, fixed parameters, distance, halo taper and backend, together with sampler settings, source hashes, git state and package versions. Reports verify the data snapshot and reproduce the saved best-sample likelihood before plotting. The driver refuses an existing run directory, including an interrupted run.

Older runs retain their original products. Reports reconstruct them from recorded options and current data, with a warning and a likelihood check. New evidence comparisons use the saved observation arrays; legacy comparisons retain conservative input hashes and the full mock provenance. See **docs/CODE_ANALYSIS_AUDIT.md** for the audit and repair record. These snapshots preserve resolved fit inputs, not a complete software installation or the raw-catalogue reduction history.

8 Cross-checks

The inference is not allowed to rest on one implementation.

A second Jeans engine. JamPy’s spherical solver runs behind the same likelihood. The mass components are expanded as projected MGEs so it can consume them, and the anisotropy family matches JamPy’s logistic parametrisation. The non-Gaussian mass-to-MGE conversion and independent quadrature make this a numerical cross-check, not an exact identity. At the K1 and cored best samples, total log likelihoods differ by about 0.7, with larger opposing HST contributions. The cause of that difference has not been isolated.

A distribution-function model. An AGAMA Cuddeford–Osipkov–Merritt DF, again behind the same likelihood. It is a *limited* cross-check and is described as such: a successful construction does not certify a non-negative distribution function, because AGAMA clips negative values to zero, which can change the realised tracer density (a ratio of 1.045 at 0.11 pc in one trial). Two further limits: its Cuddeford family cannot represent the anisotropy profile we fit, and the backend softens the point mass with a 10^{-4} pc Plummer radius, so its central limit is not Keplerian. It is available as a separate diagnostic; there is no automatic DF-positivity veto on posterior draws. Its CLI family has different anisotropy parameters and cannot replay an arbitrary Jeans posterior vector.

Published analyses. Presets reproduce the assumptions of individual papers and compare conditional results. They are approximations to published assumptions, not independent reproductions of each paper’s full data selection and dynamical method.

Injection. Earlier K1 mock fits are exploratory checks, not validation of the revised 89-point ladder. The generator draws symmetric Gaussian errors using the mean of each asymmetric error pair and clips low values. It is not an exact draw from the fitted split-normal likelihood. New mock runs record that noise prescription and restore the source run’s model, including its ladder switches. The scientific design of a rung-specific recovery programme can therefore be chosen explicitly.

9 What the model does not include

Stated plainly, because each is a route by which a spurious halo could appear.

1. **Sphericity.** ω Cen is flattened. A spherical model absorbs flattening into mass and anisotropy.
2. **Rotation is removed, not modelled.** The streaming term is subtracted from the model’s second moment. A rotating model would be preferable.
3. **Equilibrium.** The outer bins are near plausible pericentric tidal scales. Their ratio to r_J depends on the orbit and tidal-radius convention; the data are not demonstrably all inside an equilibrium region.
4. **Diagonal likelihood.** The error-model choice, the rotation-curve uncertainty and the shared field template can act coherently across bins. Rotation uncertainty enters as per-bin errors, while the raw/eta choice is tested through separate fits. A shared nuisance parameter is the correct treatment; comparing separate fits is the chosen alternative.
5. **A single tracer population.** Mass segregation makes the dispersion depend on stellar mass, and the selections differ in magnitude between instruments.

10 The runs

The default data contain 89 points. Rung 0 fixes $\beta = 0$ and removes scales.

Family	$\ln Z$	χ^2 / number of points
K1	-190.02 ± 0.48	772 / 89
K2 cored	-188.94 ± 0.36	761 / 89
K2 NFW	-190.09 ± 0.35	763 / 89

Rung 0 completes the intended baseline stage. Its weak cored preference and structured outer residuals provide a reference for the next controlled comparisons. The central fit is acceptable under the adopted errors. The component ratios motivate varying anisotropy; they do not directly measure intrinsic $\beta(r)$.

The next stage is constant β , followed by a specified turnover extension if justified. The existing monotonic form is available as an intermediate control. Instrument scales are outside the compared sequence. No rung-1 or turnover posterior on these data is complete. See `docs/MODELLING_PLAN.md` for the authoritative recipes and `docs/RUNGO_ANALYSIS.md` for the conditional numerical results.

From the repository root with `PYTHONPATH=src`, use `python -m ocen_dm.cli`, or the installed `ocen` command. The baseline requires `fit -family K1 -isotropic -no-scales`; choose a fresh `-label` to preserve old results. A bare `fit` selects the legacy varying model with scales, not the baseline.

`-gaia-errors eta` and `-gaia-rotation ours` are sensitivity variants for Gaia only. HST always uses the published rotation curve in the current loader. There is no HST rotation switch; its locally corrected means cannot replace absolute streaming. Reports preserve these choices and check the saved likelihood. Mock generation from a source run restores its model independently of the fitted family's stage. Fits are launched only when explicitly requested.

References

- An, J. H., Evans, N. W. 2006, ApJ, 642, 752. <https://ui.adsabs.harvard.edu/abs/2006ApJ...642..752A>
- Binney, J., Mamon, G. A. 1982, MNRAS, 200, 361. <https://ui.adsabs.harvard.edu/abs/1982MNRAS.200..361B>
- Vasiliev, E., Baumgardt, H. 2021, MNRAS, 505, 5978. <https://arxiv.org/abs/2102.09568>